

Can a model tell when it is wrong?

A living study of confidence signals at the MiniMax frontier

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12 July 2026

Research home Field journal Paper MiniMax protocol

Living research · preliminary · not peer reviewed

This page separates what the data already support from what is still being tested. Detailed methods live in the protocols and paper.

The short answer

Difficulty is a useful starting point. Billing and timing metadata may make that estimate better. The words in a model's reasoning trace did not.

A fresh, prospectively registered MiniMax study is now testing whether the metadata result holds up. A separate idea—launching backup models when the trace looks troubled—failed its no-spend gate and was stopped.

The problem in one minute

Suppose a model answers a hard question. We want a number such as:

There is a 72% chance this answer is correct.

A difficulty-response curve supplies the first estimate: easier tasks should receive higher confidence than harder ones. But two equally hard calls can unfold very differently. One may finish cleanly; another may consume its whole token budget, repeat itself, or backtrack for minutes.

This project asks whether those call-level signals improve the probability of correctness **for this particular answer**.

Supported — the baseline

Task difficulty gives a useful prior probability of success.

Exploratory — metadata

Token use and elapsed time predicted mistakes in old MiniMax calls. Fresh validation is under way.

Not supported — trace words

The frozen dictionary of hedging, repetition, backtracking, and give-up language did not produce a reliable confidence score.

Not supported — automatic fan-out

The frozen warning policy launched backup models on almost every call, so its paid experiment was cancelled before spending.

Censored — existing GLM calls

The 14 historical GLM-5 calls span three unpinned providers and contain only five silent errors. They cannot identify a confidence replication.

What exactly counts as failure?

The study keeps three outcomes separate:

1. **Correct completion.** The model finishes and the answer passes the task’s mechanical checker.
2. **Silent error.** The model finishes normally, but the answer is wrong.
3. **Loud failure.** The call truncates, refuses, times out, fails to parse, or returns an API error.

The confidence question compares correct completions with silent errors. Loud failures are operationally important, but they are modelled separately because they already announce themselves.

The primary score is the **Brier score**, which measures the squared error of a stated probability. A useful confidence model must improve probability calibration, not merely put failures near the top of a ranking. Repeated calls from the same generated puzzle are always kept in the same evaluation fold.

Finding 1: metadata looked promising in old calls

The retrospective MiniMax cohort contains **145 calls on 30 generated instances**:

61

correct completions

24

silent errors

60

loud failures

The selected metadata model reached **AUROC 0.852** and improved Brier score by **36.9%** over the difficulty curve alone. Its instance-bootstrap 95% interval for that improvement was 11.5% to 57.6%.

In plain language: silently wrong completions used about **54,700 billed tokens**, compared with **36,100** for correct completions, and took about **756 seconds**, compared with **524 seconds**. Their effective billed token rates were almost identical. MiniMax did not appear to generate more slowly when it was wrong; it simply continued for longer.

The trace-only model was much weaker: **AUROC 0.671** and **7.4% Brier skill**, with an uncertainty interval that crossed zero. The earlier trace-ranking result did not transfer into a dependable probability of correctness.

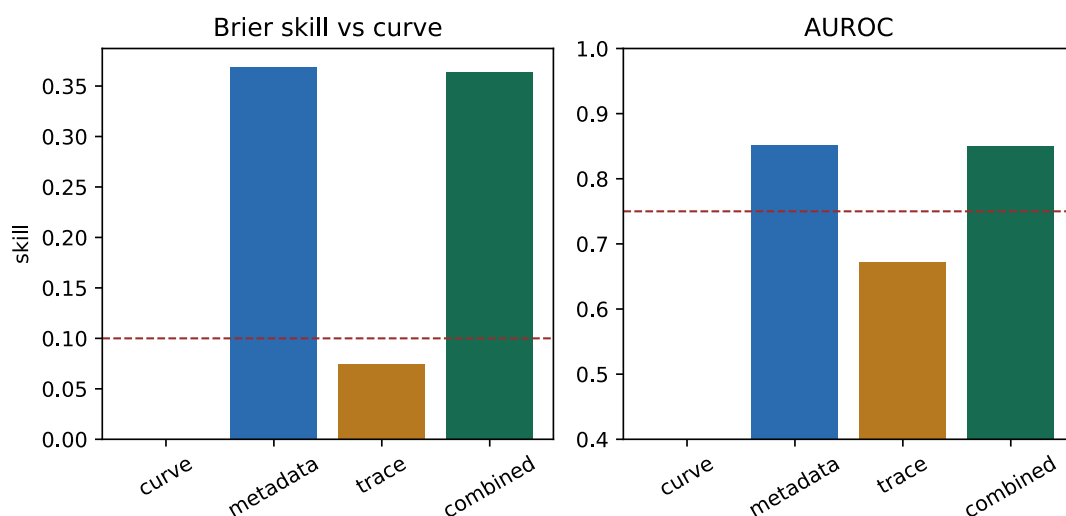


Figure 1: Four retrospective confidence models. Metadata and the combined model clear the exploratory thresholds; trace features alone do not.

What this result does—and does not—say

It says metadata is worth a fresh test. It does **not** yet show that the score will work on new calls. The old calls were used to choose the model, so this finding remains **Exploratory**.

Study 1: the fresh MiniMax test is running

The prospective test was frozen before collecting its evaluation calls:

- one model: or/minimax-m2.5;
- one route: openrouter/Parasail;
- a fixed 65,536-token cap and prompt version;
- a metadata-only confidence model selected in advance;
- one final analysis, with no peeking and retuning;
- stop after at least 60 correct and 60 silently wrong completions, 600 calls, or USD 40—whichever comes first.

The result will count as supported only if Brier skill remains at least 10%, its instance-bootstrap 95% interval excludes zero, and AUROC remains at least 0.75. If it fails, the negative result is the result.

[Read the frozen MiniMax protocol](#) · [Download the exploratory result JSON](#) · [Download the compact study table](#)

GLM replication audit: the old calls are not enough

The released database contains **14 calls on z-ai/glm-5**: nine correct completions and five silent errors. That is useful reconnaissance, but not a replication of the MiniMax confidence result.

Three problems prevent a calibrated comparison:

1. The cohort has five silent errors, below the registered minimum of 20.
2. Calls were served unpinned across Phala, SiliconFlow, and Z.ai routes.

3. Correct and wrong calls occupy different difficulty cells; no tested level contains both outcomes, so difficulty and metadata effects are confounded.

The audit is therefore **Censored**, not negative. It says the available data cannot answer the question; it does not say GLM lacks a confidence signal. No new calls were made and new spend was USD 0.

Download the GLM cohort audit

Study 2: the trace-triggered backup idea failed

The control idea was simple:

1. Start with a trace-visible primary model.
2. Watch for persistent backtracking, hedging, repetition, or give-up language.
3. If risk stays high, pause external side effects and launch challenger models.
4. Accept only an answer that passes an external verifier; cancel the rest.

This differs from a conventional model council. A council launches several models by default and combines completed answers. The proposed controller was supposed to launch backups only when needed.

Before making any paid calls, the frozen trigger was replayed over every historical call with a visible reasoning trace.

84 / 89

calls launched backups

50 / 50

failures triggered

34 / 39

correct calls also triggered

USD 0

new API spend

The policy caught every failure, but only because it escalated **94.4% of all calls**. Its false-hedge rate on correct completions was **87.2%**, far above the registered 40% ceiling. A simple timer matched to the same launch rate also caught every failure.

Changing the replay observation size did not rescue the policy. Coarser chunks reduced false alarms, but also lost failures; every tested setting failed the gate.

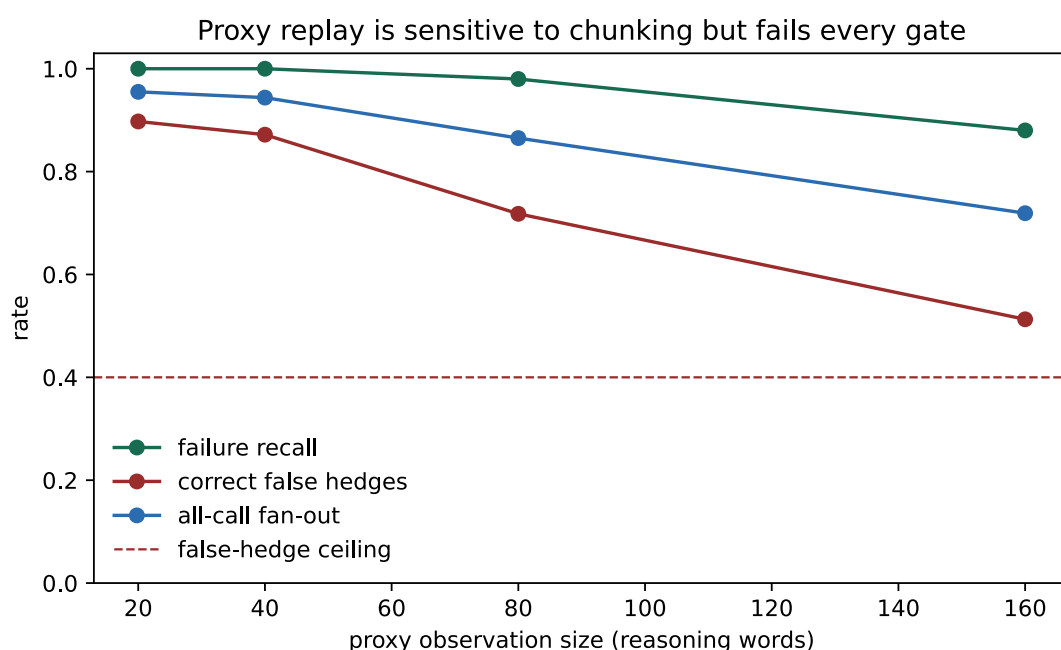


Figure 2: Failure recall stays high as replay chunks grow, but false hedges never fall below the registered ceiling.

The historical traces lack timestamped chunks, so the promised 15-second warning-time test is **Censored**. That missing clock does not change the decision: the false-alarm gate already failed. The live branch was stopped and the threshold was not retuned.

Read the frozen design and outcome · Download the replay result · Download per-call replay decisions

What to take away today

Supported

Difficulty-response curves provide a useful baseline probability.

Exploratory

Completion length and elapsed time may improve per-answer confidence for MiniMax. The prospective test is still running.

Not supported

The frozen trace dictionary is not a strong calibrated confidence model, and the frozen trace trigger is not a selective backup policy.

Censored

End-to-end billed tokens per second is not generation throughput. Queueing, retries, and transport are mixed into the denominator; streamed timing is needed to measure actual output rate. The existing GLM-5 pilot is also too small and route-confounded to serve as a confidence replication.

The open record

This is intentionally a living research page rather than a polished preprint. The field journal preserves the sequence of questions, mistakes, protocol freezes, and negative results. The

paper contains the broader difficulty-curve work. A conventional preprint will wait for at least one prospectively validated result.

The programme connects generated evaluation and dynamic testing (GSM-Symbolic, Dynamic Evaluation) with difficulty-aware routing (IRT-Router, RouteLLM) and answer agreement (Self-Consistency). Its narrower question is whether signals produced during one model call help us trust that call.

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